
Making a Cellular Potts Model of Collective Migration

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1 Introduction

Active cell migration is essential in numerous biological processes, such as wound healing and immune responses. These processes often occur in complex and dense environments. A prime example discussed in class is the T cell, which is capable of navigating smoothly through these environments.

To better study these movements, we use a Cellular Potts Model (CPM) to simulate collective cell migration. Whereas the lecture studied cell migration in a single-lane traffic scenario, this report will analyze the impact of static round obstacles on cell migration dynamics in a crowded grid.

The simulation is based on Act-CPM [1], where cell movement is driven by the polymerization of actin fibers at the cell front, therefore creating directed motion through a positive feedback loop. Consequently, cells do not require external forces to interact, allowing for spontaneous complex collective behavior.

In the following sections, we describe the implementation of our CPM experiment and the specific placement of obstacles. We then present an analysis of the resulting cell dynamics, concluding with an interpretation of how these obstacles alter the collective movement of cells.

2 Methods

2.1 Act-CPM

The Act-CPM models cell movement as a series of stochastic pixel updates, where the probability of an update is characterized by the minimization of the configuration energy ($\mathcal{H}$).

$$\mathcal{H} = \sum_{(i,j) \in \mathbf{E}} J_{\tau_i, \tau_j} (1 - \delta_{ij}) + \sum_{\sigma} \lambda_{V, \tau} \cdot (V_{\sigma}(t) - V_{\tau})^2 + \sum_{\sigma} \lambda_{P, \tau} \cdot (P_{\sigma}(t) - P_{\tau})^2$$

Here, the first term accounts for adhesion between cells or the background, where J is a penalty term. The second and third terms act as constraints, penalizing deviations from the cell's target volume (V_{τ}) and target perimeter (P_{τ}) with weights scaled by λ_V and λ_P , respectively.

To simulate movement, the model evaluates a "copy attempt" where a pixel at a source position p_s tries to displace a pixel at target position p_t . The success of this attempt depends on the total change in energy, $\Delta\mathcal{H}$, which in the Act-CPM combines standard CPM physics with an additional activity-based term:

$$\Delta\mathcal{H} = (\mathcal{H}_{\text{new}} - \mathcal{H}_{\text{old}}) + \Delta\mathcal{H}_{\text{act}}(p_s \rightarrow p_t)$$

This activity term, $\Delta\mathcal{H}_{\text{act}}$, represents the biological force of actin polymerization. It biases movement by looking at the "memory" of recent activity within the cell. The contribution is calculated as:

$$\Delta\mathcal{H}_{\text{act}}(p_s \rightarrow p_t) = -\frac{\lambda_{\text{act}}}{\text{Max}_{\text{act}}} (\text{GM}_{\text{act}}(p_s) - \text{GM}_{\text{act}}(p_t))$$

In this expression, $GM_{\text{act}}(p)$ is the geometric mean of the activity values in the neighborhood of pixel p . By subtracting the activity of the target from the source, the model creates a positive feedback loop: pixels with high activity (recent updates) are more likely to "push" into their neighbors.

2.2 Obstacle

To model obstacles, we introduced an additional, non-motile cell kind that is visually distinguished by color. In contrast to the motile cells, obstacle cells are initialized to be half the size as their target volume is set to 250 pixels. During initialization, each obstacle is seeded directly as a filled disk whose radius is computed from its target volume, ensuring that obstacles start close to their intended size rather than growing from a single pixel.

To keep obstacles stationary, we disabled activity driving motion by setting the activity term to zero for this cell kind. At the same time, a strong volume constraint together with a perimeter constraint consistent with a circular shape ensures that obstacles remain compact and largely undeformed throughout the simulation. With this configuration, obstacle cells behave as static, space-occupying objects around which motile cells can move.

2.3 Obstacle Grid

Obstacles are placed on the grid in a regular lattice by overriding the simulation's `initializeGrid` method. First, a `GridManipulator` is added to allow explicit control over cell seeding and pixel assignment. Obstacle positions are then iterated over using a fixed spacing and margin, where the spacing parameter directly determines obstacle density.

At each lattice position, a new obstacle cell is created using `seedCellAt`, which returns a unique cell identifier. A circular set of pixels with the appropriate radius is then generated using `makeCircle`, and these pixels are assigned to the obstacle cell with `assignCellPixels`. This procedure results in a uniform grid of half-sized, disk-shaped obstacle cells.

Once the obstacle grid is initialized, motile cells are seeded randomly on background pixels only, ensuring that they do not overlap with obstacles at the start of the simulation. Random initialization avoids introducing artificial spatial order among motile cells and allows the observed collective migration patterns to emerge from cell-cell interactions and obstacle density alone. This setup enables a systematic investigation of how regularly spaced obstacles affect collective cell migration while keeping all other aspects of the simulation unchanged.

3 Results

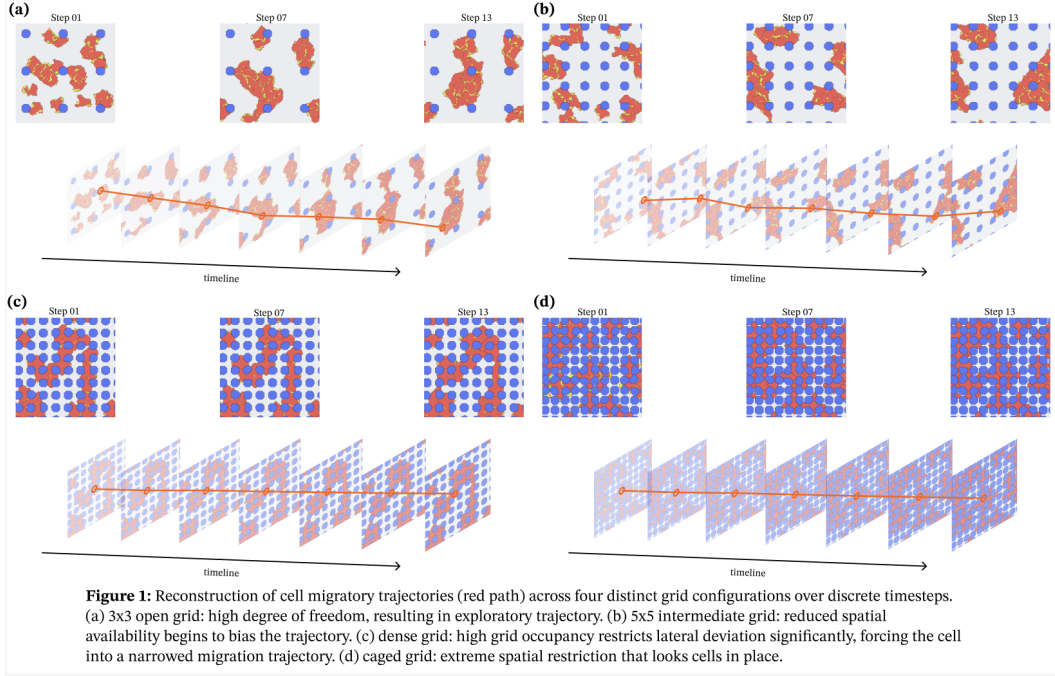
3.1 Impact of Obstacle Spatial Arrangement

The spatial arrangement of obstacles critically determines whether cells are able to migrate or become mechanically arrested. When the gap between neighboring obstacles is small, cells that spawned inside the grid become almost immovable. In this configuration, the available free space is insufficient to accommodate cell deformation, and migration is effectively suppressed. Cells inside the grid display minimal displacement over time. ¹

Obstacles act as a mechanical filter that regulates, rather than prevents, cell infiltration. Cells continue to enter the grid, but the ease of entry is governed by a gradient of resistance defined by the grid's density. This transition is controlled primarily by the perimeter penalty term: as a cell squeezes through narrow gaps, it must increase its surface-to-volume ratio, incurring an energetic cost.

The likelihood and rate of entry are therefore determined by the balance between the cell's motile drive and this geometric constraint. In highly dense grids, the perimeter penalty becomes a limiting factor, slowing the infiltration process as cells must undergo more extreme deformations. Instead of a total exclusion, the grid imposes a "selection pressure" where entry is still observed, but its efficiency is scaled by the physical dimensions of the environment.

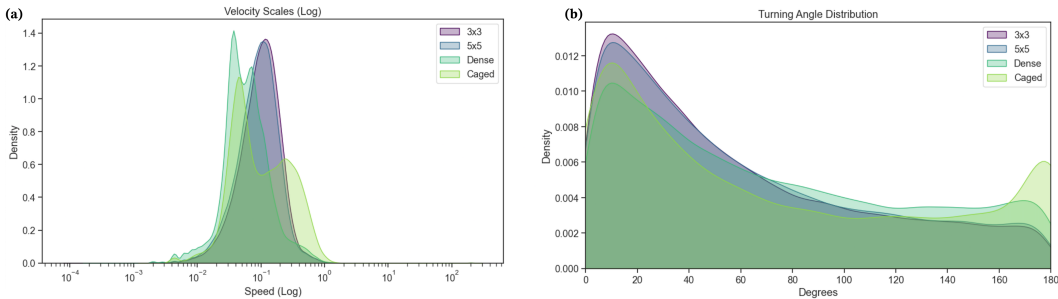
When obstacle spacing is increased or when cells are forced into the grid, a qualitative change in behavior emerges. Rather than moving independently, cells tend to migrate collectively. Typically, a single cell enters first and effectively acts as a pathfinder, deforming to pass through the construction. Following cells go along the same path, benefiting from the locally altered configuration and reduced resistance. This leads to chain-like or group-based entry into the grid rather than isolated single-cell migration.



3.2 Quantitative Analysis of Cell Motion

Velocity distributions extracted from centroid motion show a clear dependence on obstacle density. 2Denser grids are associated with systematically lower velocities, while sparser grids exhibit higher characteristic speeds. In particular, cells in the locked or caged configuration display the lowest velocities, confirming strong mechanical confinement. The 3x3 and 5x5 grids show intermediate behavior, with the 3x3 configuration allowing higher velocities than the 5x5 grid, consistent with increased available space.

Turning angle distributions further support this interpretation. In sparser grids, turning angles are biased toward smaller values, indicating more persistent and directed motion. As grid density increases, the turning angle distribution broadens, with a higher probability of large reorientation angles. This reflects frequent directional changes induced by collisions with obstacles and neighboring cells. In the densest and caged configurations, turning angles approach a near-uniform distribution, showing abundance of motion dominated by local rearrangements rather than directed migration.



3.3 Overall Effect of Static Obstacles on Collective Migration

Taken together, the centroid-based velocity and turning analyses demonstrate that static obstacles incentivize collective migration by simultaneously limiting speed and disrupting directional persistence.

Increasing obstacle density progressively shifts the system from directed collective migration toward constrained, low-velocity, and highly reoriented motion. In the extreme case, obstacles induce a locked state where cells are immobilized.

These results confirm that static obstacles do not simply reduce motility but actively reshape migration modes, promoting collective path-following in sparse environments and immobilization in dense ones. The observed transitions arise from geometric constraints imposed by the environment rather than changes in motility parameters.

4 Discussion and Conclusion

Several limitations of this work should be noted. First, the observed behaviors are dependent on the chosen model parameters 1, including volume constraints and interaction strengths. While some observed results were robust across the tested conditions, different parameter values may shift the outcome. A sensitivity analysis would be required to quantify the effectiveness of these transitions.

Second, the obstacle configurations considered here represent a limited set of regular grid geometries. Although clear differences were observed between sparse, intermediate, dense, and caged arrangements, other spatial organizations or irregular obstacle layouts may lead to different migration patterns. The results should therefore be interpreted as geometry-dependent rather than universal.

Finally, motion was characterized using centroid-based metrics, which capture collective displacement and reorientation but do not fully resolve local deformation or force distributions at cell–cell and cell–obstacle contacts. While sufficient for distinguishing between different migration states, additional metrics would be needed to link these behaviors more directly to mechanical stress or shape changes.

In conclusion, this work shows that static obstacles act not merely as barriers but as active regulators of collective migration. By tuning obstacle spacing and density, the system transitions between free migration, collective path-following, and mechanical trapping. These results showcase the central role of environmental geometry in shaping collective cell dynamics and provide a framework for interpreting migration patterns in confined and structured environments.

References

- [1] I. Niculescu, J. Textor, and R. J. de Boer. Crawling and gliding: A computational model for shape-driven cell migration. *PLoS Comput. Biol.*, 11(10):e1004280, Oct. 2015.

A Code

```

let OBSTACLE_SPACING = 30
let OBSTACLE_MARGIN = 40
function makeCellOfKind(kind){
  if (sim.C.makeNewCellID.length >= 1){
    return sim.C.makeNewCellID(kind)
  }
  const cid = sim.C.makeNewCellID()
  sim.C.setCellKind(cid, kind)
  return cid
}

function seedDisk(cid, cx, cy, r){
  const r2 = r*r
  for(let dx = -r; dx <= r; dx++){
    for(let dy = -r; dy <= r; dy++){
      if(dx*dx + dy*dy <= r2){
        const x = cx + dx
        const y = cy + dy
        if(x >= 0 && x < config.field_size[0] && y >= 0
           && y < config.field_size[1]){
          sim.C.setpix([x,y], cid)
        }
      }
    }
  }
}

function seedObstaclesRegular(spacing=50, margin=50, radius=6){
  for(let x = margin; x < config.field_size[0] - margin; x += spacing){
    for(let y = margin; y < config.field_size[1] - margin; y += spacing){
      const cid = makeCellOfKind(2)
      seedDisk(cid, x, y, radius)
    }
  }
}

```

Listing 1: Obstacle seeding functions

B Model Parameters

Parameter	Symbol	Cell ($\sigma = 1$)	Obstacle ($\sigma = 2$)
Target Volume	V	500	250
Volume Elasticity	λ_V	50	500
Target Perimeter	P	340	≈ 39.6
Perimeter Penalty	λ_P	2	50
Max Activity	Max_{act}	20	0
Activity Lambda	λ_{act}	200	0
Adhesion (to Medium)	$J(\tau, 0)$	20	30
Adhesion (Cell-Obstacle)	$J(1, 2)$	30	

Table 1: Model parameters and adhesion coefficients. The J values define the surface tension between the cells, the obstacles, and the medium (0), while λ_P regulates the deformation cost during grid entry.